

SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

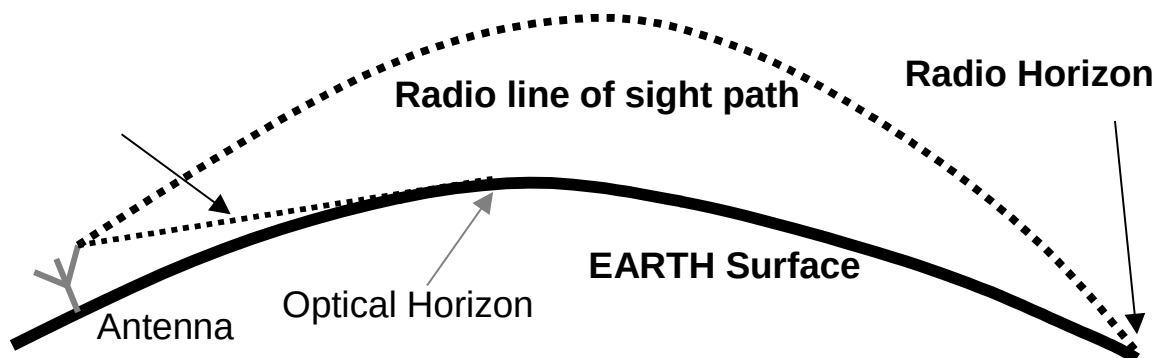
Wireless Communication & Networking (EGE321)

Tutorial 4: Propagation Path-Loss Model

- 1) *With the aid of diagram, distinguish the differences between optical line of sight and radio line of sight.*

Optical line of sight is the direct free-space distance that exists between two end-points. It is the **distance seen by a distant transmitter to the distant receiver**. (“face to face”) To have a clear line of sight, there must be no obstructions between the two end-locations.

However due to the density of the atmosphere changing less dense as the height increases, **radio signals are refracted into slightly curved paths**, and hence communication antennas can “see” each other even when they are beyond the visible horizon. This is the radio horizon – **the distance that is seen by the radio signal**. Radio Horizon is usually taken as 1.33 times longer than the optical distance.



- 2) *Describe briefly the 3 basic mechanisms listed below that occur in radio wave propagation.*

a) Reflection

Radio waves may be reflected from various substances or objects they meet during propagation if the dimension of the object is **very large compared to the wavelength of the radio wave**. The amount of reflection depends on the reflecting material.

When radio waves are reflected, a **phase shift will normally occur**. Radio waves that keep their phase relationships (**in-phase**) after reflection normally **produce a stronger signal** at the receiving site. Those that received **out of phase** produce a **weaker signal** - **fading**. In the extreme, the signals could just cancel each other.

b) Diffraction

When radio waves meet a **sharp** obstacle, they have a natural tendency to **bend around the tip of the obstacle**. This is known as diffraction. This **results in a change of direction of part of the wave energy from the normal line-of-sight path**. Because of this change, it is **possible to receive energy around the edges of an obstacle** or at some distances below the highest point of an obstruction in the shadow zone. This diffracted radio frequency energy is weaker and may still be detected by a receiver.

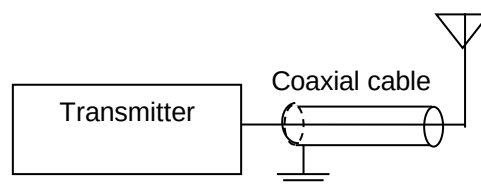
c) Scattering

When the radio waves meet a rough surface or very small objects whose **dimensions are comparable to the radio wave** such as dust particles, leaves or raindrops, it will tend to be **reflected into many directions (scattered)**. This scattering effect is employed by **weather radar** to detect the presence of rain nearby. The effect is best observed at around 10GHz.

3) Define the concept of Free Space Propagation.

Free space assumes a channel **free of all hindrances or obstacles to RF propagation such as absorption, reflection, refraction or diffraction**. The energy arriving at the receiver is assumed to be **only a function of the distance from the transmitter (following the inverse square law)**. A free space channel characterizes **an ideal RF propagation path**.

4) A transmitter transmits a signal with a wavelength of 15m over the air.



a) Determine the frequency of the signal.

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{15} = 20 \text{ MHz}$$

b) If the signal travels through the coaxial cable with a speed of $2 \times 10^8 \text{ m/s}$, calculate the wavelength of the signal in the coaxial cable.

$$\lambda = \frac{v}{f} = \frac{2 \times 10^8}{20 \times 10^6} = 10 \text{ m}$$

- 5) A satellite that is 500km above the earth transmits a signal at 1.6GHz to an earth station. The transmit power of the satellite is 100W. Estimate the received signal in dBm at the antenna of the earth station. Assume free space path loss.

$$\begin{aligned}\text{Free space loss } L \text{ (dB)} &= 32.4 + 20 \log d \text{ (km)} + 20 \log f \text{ (MHz)} \\ &= 32.4 + 20 \log 500 + 20 \log 1600 \\ &= 150.5 \text{ dB}\end{aligned}$$

$$\begin{aligned}P_T &= 100 \text{ W} &&= 50 \text{ dBm.} \\ P_R \text{ (dBm)} &= P_T \text{ (dBm)} - L \text{ (dB)} \\ &= 50 - 150.5 \\ &= \mathbf{-100.5 \text{ dBm}}\end{aligned}$$

- 6) Calculate the EIRP of a microwave transmitting installation where the transmitter output power is 0.5 W, the total feeder cable loss is 5 dB, and the transmitting antenna gain is 33 dB.

$$\begin{aligned}\text{EIRP (dBm)} &= P_T \text{ (dBm)} - L_f \text{ (dB)} + G_T \text{ (dB)} \\ &= 10 \log (0.5 \text{ W} / 0.001 \text{ W}) - 5 \text{ dB} + 33 \text{ dB} \\ &= 26.99 \text{ dBm} - 5 \text{ dB} + 33 \text{ dB} \\ &= 55 \text{ dBm}\end{aligned}$$

- 7) A microwave carrier system is operating at 450 MHz and has a range of 25 km. The transmitter power is 25 W and the receiving and transmitting antenna gains are 24 dB each. Assuming no other losses, calculate the following the free space loss.

$$\begin{aligned}\text{Free space path loss } L_P \text{ (dB)} &= 32.44 + 20 \log d \text{ (km)} + 20 \log f \text{ (MHz)} \\ &= 32.44 + 20 \log 25 + 20 \log 450 \\ &= 113.46 \text{ dB}\end{aligned}$$

- 8) An aircraft is now approaching landing at a height of 1000m and is transmitting 5W to a control tower 10km away. The height of the control tower is 90m. Assume that there is a **ground reflected signal** to the control tower. Determine the received power in dBm at the control tower.

Since there is a ground reflected wave, we use the **Plane earth model**.

$$\begin{aligned}L \text{ (dB)} &= 120 + 40 \log d_{km} - 20 \log [h_r \text{ (m)} h_t \text{ (m)}] \\ &= 120 + 40 \log 10 - 20 \log (1000 \times 90) \\ &= 60.9 \text{ dB}\end{aligned}$$

$$\begin{aligned}P_T &= 5 \text{ W} &&= 37 \text{ dBm.} \\ P_R \text{ (dBm)} &= P_T \text{ (dBm)} - L \text{ (dB)} \\ &= 37 - 60.9 \\ &= \mathbf{-23.9 \text{ dBm}}\end{aligned}$$